

Cohort Studies

Classify a healthy population by exposure, follow it forward in time, and let the disease reveal itself — the design behind Snow's cholera map, the British Doctors Study, and today's million-person biobanks.

Based on the chapter by Pascal Wild, Anthony B. Miller, David C. Goff Jr. & Karin Bammann · *Handbook of Epidemiology*

ROADMAP

From Assembling a Cohort to Reading Its Results

- 1 A century of cohorts** — from a London cholera map and pellagra kitchens to today's million-person biobanks.
- 2 Two ways to build one** — a concurrent study run forward from today, or a historical one built from records already on file.
- 3 Counting disease properly** — the shift from simple risk to person-time incidence rates, and what “at risk” really means.
- 4 Comparing outward and inward** — checking the cohort against the general population, and the bias that comparison can hide.
- 5 Measuring an effect's size** — risk ratios, rate ratios, and the attributable-risk family that turns association into public-health impact.
- 6 Modeling exposure that keeps changing** — person-time sliced by dose, duration, and lag, and the regression models built for it.
- 7 Choosing a population, measuring exposure** — who gets in, how exposure is captured, and the confounders that complicate both.
- 8 Outcomes, follow-up, and ethics** — ascertainment bias, losses that bias results, and consent for a purpose not yet known.

THE BASIC DESIGN

What Is a Cohort Study?

A cohort study starts with a defined, disease-free population. Its members are classified by whether they carry a risk factor or exposure of interest, then followed forward in time to see who develops the disease (or dies) and who does not. Comparing the exposed group's experience to the unexposed group's is what answers the question.

Cohort vs. case-control

Because a cohort study starts from exposure and watches for whatever follows, it can relate many different diseases to one exposure at once. Its classical counterpart, the case-control study, runs in the opposite direction: it starts from disease and looks backward at exposure. The trade-off is that cohort studies are often restricted to a limited number of exposures and confounders that can practically be measured, especially when historical records are the only source.

A CENTURY OF COHORTS

A Design That Has Outlived Every Rival Hypothesis

For over a century, cohort studies have been epidemiology's way of watching a hypothesis play out in a real population — across an extraordinary range of subject matter: infections, nutrition, occupational hazards, and everyday lifestyle habits.

1849

Snow maps cholera deaths by water company

Early 1900s

Goldberger traces pellagra to diet, not infection

Mid-late 1900s

Occupational and lifestyle cohorts reshape public health policy

2000s–today

Million-person biobanks probe genes, environment, and COVID-19

1849 • LONDON

Snow's Cholera Map Is a Cohort Study

Two companies supplied drinking water to overlapping London neighborhoods. In 1849 the Lambeth company moved its water intake upstream of London's sewage discharge; the Southwark & Vauxhall company kept drawing water downstream of it. John Snow classified households by which company served them — an exposure fixed before anyone fell ill — then counted cholera deaths in each group.

Lambeth (intake moved upstream)

37 deaths

per 10,000 households

Southwark & Vauxhall (still downstream)

315 deaths

per 10,000 households

An early cohort study on an infectious disease, decades before the germ theory of cholera was accepted.

EARLY 1900S · U.S. SOUTH

Goldberger and the Pellagra Puzzle

Pellagra was a systemic disease endemic across the southeastern United States, and many suspected it was infectious. Joseph Goldberger instead examined households' dietary exposures and found a cornmeal-subsistence diet closely tied to who developed the disease.

Ruled out

Person-to-person transmission — subsequent trials showed pellagra could not spread the way an infection would.

Confirmed

The disease could be prevented by a “pellagra-preventive factor” — later identified as niacin.

A cohort approach settled a live scientific dispute — infectious cause or nutritional deficiency — well before the responsible vitamin had a name.

INFECTIOUS DISEASE, CONTINUED

From HIV to SARS-CoV-2

Cohort studies remain a central tool for infectious disease. Decades before COVID-19, McCray (1986) followed a cohort of healthcare workers exposed to blood and body fluids to quantify their risk of developing AIDS.

Wrtil et al. (2021) followed a cohort of blood donors to test whether pre-existing immunity to seasonal coronaviruses affected susceptibility to SARS-CoV-2.

The U.S. NIH RECOVER cohorts and the Collaborative Cohort of Cohorts for COVID-19 Research were established to generate new knowledge on incidence, susceptibility, and post-acute sequelae — long COVID.

OCCUPATIONAL EPIDEMIOLOGY

Cohorts That Turned Suspicion Into Policy

Workers exposed to a specific substance, compared against the rest of the industry or the general population, have repeatedly confirmed hazards first suspected from clinical observation or animal studies.

Dyes

Bladder cancer (Case et al. 1954)

Mustard gas

Respiratory cancer (Wada et al. 1968)

Benzene

Leukemia (Rinsky et al. 1987)

Asbestos

Mesothelioma (Liddell 1997; Selikoff 1965)

Ulvestad (2004) followed Norwegian insulation workers hired between 1930 and 1975 through to 2002, and found their mesothelioma and lung cancer risk clearly elevated over the general population.

LIFESTYLE EXPOSURES

Everyday Habits, Tracked for Decades

Sitting vs. climbing stairs

Morris et al. (1953) found British bus drivers had roughly twice the heart disease risk of the more active conductors, who climbed the stairs collecting tickets — confirmed comparing sedentary postmen to telephonists and clerks.

Doctors and cigarettes

Doll and Hill began a 1951 cohort of British physicians using a mailed smoking questionnaire. Doll and Peto (1976) later showed a tenfold increase in lung cancer deaths among smokers versus nonsmokers.

Both results held up because exposure — activity level, smoking habit — was recorded before disease struck: the cohort design's signature advantage.

A CAUTIONARY TALE

Alcohol, Mortality, and the “Sick Quitter”

Doll et al. (1994a) found a U-shaped relationship between alcohol and mortality among British doctors: both abstainers and heavy drinkers had higher mortality than moderate drinkers (one to two drinks a day). One objection: perhaps some “abstainers” were really former drinkers who had already quit because their health was failing — inflating the abstainer group’s apparent risk.

Eigenbrodt et al. (2001), using the Atherosclerosis Risk in Communities (ARIC) study, tracked perceived health status over time and found that declining health **preceded** people stopping drinking — supporting the concern that lumping former drinkers in with lifelong abstainers may have inflated the abstainer group’s apparent risk.

A reminder that a cohort’s exposure categories are only as trustworthy as the history behind them.

TODAY

Million-Person Cohorts

The largest cohorts now underway dwarf anything Snow or Goldberger could have imagined, combining genetic and environmental data at a scale meant to power research for decades.

All of Us (NIH)

A 1,000,000-person U.S. cohort examining genetic and environmental influences on common diseases.

OMEGA-NET

A European network organizing occupational cohorts covering roughly 30 million people.

Constances, NAKO, UK Biobank, Lifelines

National cohorts in France, Germany, Britain, and the Netherlands, each enrolling several hundred thousand volunteers.

Large samples let investigators study genetic and environmental effects, and their interactions, both in the general population and within specific subgroups of interest.

THE CENTRAL IDEA

One Design, Four Moving Parts

1

Define a healthy, disease-free population

2

Classify members: exposed or unexposed

3

Follow both groups forward in time

4

Compare who develops disease, and who does not

That comparison — incidence among the exposed versus incidence among the unexposed — is the entire cohort study. Everything else in this guide is about doing each of these four steps carefully enough to trust the answer.

DESIGN TYPES

Concurrent and Historical Cohorts

Concurrent (prospective)

A defined population is assembled and screened now to exclude existing disease. Exposure, confounders, and other factors are gathered now, and the cohort is followed into the future for outcomes.

Nonconcurrent (historical)

A population is assembled from historical records, such as company files. Preexisting disease, exposure, and confounders are all assessed from data already on file, and outcomes are traced through existing records from a point in the past to the present.

A combined, “ambispective” approach is also possible (Kleinbaum et al. 1982): a cohort assembled and followed initially through historical sources, then followed onward using concurrent methods.

DESIGN TYPES

Why Accept the Harder, Historical Design?

Investigators have less control over data collection in a historical cohort — they must rely on records kept for reasons that had nothing to do with research. That disadvantage is worth accepting when the exposure meets one or more of these conditions:

It can be attributed from individual job records or similar employment data.

It is relatively rare outside the occupation or context of interest.

The induction period is long.

The health concern is substantial — an early answer has real public-health value.

Real exposures are rarely fixed, dichotomous, or constant, and cohorts are dynamic — people enter and leave the population at risk throughout follow-up. Both problems demand more advanced methods, covered later in this guide.

DESIGN TYPES

Two Modern Hybrids

Two extensions combine nearly all the statistical power of a classic cohort study with much of the cost-efficiency of a case-control study, by collecting detailed exposure information — questionnaires, biomarkers — from only a carefully chosen subsample rather than the whole cohort.

Nested case-control

Sample a handful of controls for each case from within the cohort itself. Covered in detail later in this guide.

Case-cohort

Compare all cases to a single random subcohort sampled once, rather than a fresh control set per case.

Both designs are treated in full elsewhere in the Handbook (“Modern Epidemiological Study Designs”); this guide returns to the nested case-control idea once the machinery behind it has been introduced.

COUNTING DISEASE

From Counting People to Counting Person-Time

A cohort allows direct comparison of exposed and unexposed groups, across several possible measures of effect. But analyzing cohort data takes real care, especially in one necessary change of perspective: from persons at risk to person-time at risk. Getting this shift right is what keeps the results unbiased.

The next several ideas — risk, incidence rate, “at risk” versus “under observation” — all exist to answer one question precisely: how do you fairly count disease in a group of people who were not all followed for the same length of time?

MEASURING DISEASE

The Simplest Measure: Risk

Group	Disease contracted	No disease	Total
Exposed	a	b	a + b
Unexposed	c	d	c + d
Total	a + c	b + d	a + b + c + d = N

Risk = number of new cases ÷ number of people at risk. Among the exposed, that is $a \div (a + b)$; among the unexposed, $c \div (c + d)$.

Risk is unit-free — a plain proportion, not a rate — and it assumes a fixed cohort with a stated follow-up period. Comparing cumulative incidence across studies with different follow-up lengths is a genuine fallacy to avoid.

MEASURING DISEASE

Rates, for Diseases That Take Years

Risk works cleanly for outbreaks with a short, fixed follow-up. For chronic diseases with long, uneven follow-up — where cohort members are tracked for very different lengths of time — cumulative incidence is no longer appropriate.

Incidence rate = number of new cases ÷ cumulative person-time at risk. Each person contributes time from entry until either disease develops or follow-up ends. Also called incidence density, or hazard rate.

Risk

A unit-free proportion. Ignores how long each person was actually followed.

Incidence rate

A true rate, with units of events per person-time. Accounts fairly for unequal follow-up.

Incidence rate is not the same thing as cumulative incidence — a common and consequential mix-up.

MEASURING DISEASE

What “At Risk” Really Means

At risk

A person is “at risk” at a given moment only if the event of interest could still happen to them. Someone who has had a thyroidectomy is no longer at risk of thyroid cancer; nobody is at risk after they have died. Risk depends only on the endpoint being studied.

Under observation

Whether someone is under observation depends on the cohort’s follow-up method. If their disease status becomes untraceable — say, they move abroad — they become “lost to follow-up,” or censored, from that moment on.

A subject contributes person-time to a study at a given moment if, and only if, they are **both** at risk **and** under observation at that moment.

A WORKED EXAMPLE

Ten People, One Fictitious Cohort

No.	Age at entry	Years followed	Age at end	Age at diagnosis	Person-years at risk
1	34	15	49	—	15
2	39	20	59	54	15
3	31	12	43	—	12
4	36	17	53	41	5
5	38	9	47	—	9
6	38	16	54	51	13
7	41	11	52	—	11
8	32	20	52	—	20
9	39	18	57	—	18
10	42	17	59	—	17
Total	—	155	—	3 cases	135

Three cases occurred during 155 person-years of observed follow-up, but only 135 of those years counted as time **at risk** — the other 20 years accrued after diagnosis, once each case was no longer at risk. Crude incidence rate: $3 \div 135 \approx 0.022$ cases per person-year, commonly rescaled to about 2,222 cases per 100,000 person-years.

MEASURING DISEASE

One Person Can Belong to Two Age Groups

Take cohort member No. 4: entering the study at age 36 and diagnosed at 41, they age through two brackets before diagnosis. Their five disease-free years must be split between both brackets’ incidence rates, not assigned entirely to one.

Ages 30–39: 3.5 years

Counted in the 30–39 rate’s denominator only — no case yet.

Ages 40–49: 1.5 years

Counted in the 40–49 rate’s denominator, and their diagnosis becomes that rate’s one numerator case.

Age group	Incident cases	Disease-free years	Incidence rate
30–39	0	29	0
40–49	1	78	0.013
50–59	2	28	0.071

EXTERNAL COMPARISONS

Checking the Cohort Against the Outside World

Comparing a cohort's own rates to those of an external population — usually the general population — is one important form of validation. It puts the cohort's findings in a broader context, regardless of whether an internal comparison group also exists.

Two statistical routes do this: **direct standardization**, which reweights the cohort's own rates, and **indirect standardization** — the SMR — which applies an external population's rates to the cohort. Cumulative comparisons of either kind can mask real differences, such as an unusually high rate among younger cohort members.

EXTERNAL COMPARISONS

Direct Standardization

Calculate the cohort's own age-specific incidence rates, then reweight them using the age distribution of an external, standard population. The result is a single rate showing what the cohort's experience would look like if it had the standard population's age structure.

Directly standardized rate = a weighted average of the cohort's own age-group rates, weighted by the standard population's age distribution.

Useful for comparing rates across different cohorts, or across time, on equal footing — but it performs poorly in small cohorts, where age-specific rates can rest on very few people.

EXTERNAL COMPARISONS

Indirect Standardization: the SMR

Indirect standardization applies a reference population's age-specific rates to the cohort's own age structure, producing an **expected** number of events to compare against what was actually **observed**. Comparing expected to observed deaths gives the SMR; the same logic applied to disease incidence gives the SIR (standardized incidence ratio) — SMR is used more often simply because population mortality data are more routinely available than morbidity data.

SMR (standardized mortality ratio) = observed deaths ÷ expected deaths. An SMR above 1 means greater mortality than the reference population; below 1 means less.

Statistical tests and confidence intervals exist for a single SMR, built from the Poisson distribution of the observed count. Direct standardization's poor behavior in small cohorts is exactly why indirect standardization is usually preferred in practice.

EXTERNAL COMPARISONS

Can the SMR Be Read as a Risk Ratio?

If a cohort was assembled to study one specific risk factor — an occupational exposure, say — it is tempting to read the SMR itself as the risk ratio due to that factor.

That interpretation is only valid if the cohort is comparable to the general population on **every other** factor besides the one under investigation. The chapter is blunt about this: it is “obviously only rarely the case.”

What usually breaks the comparison? The next slide names the biggest culprit.

EXTERNAL COMPARISONS

The Healthy Worker Effect

The general population includes the very ill and the very poor, who are rarely represented in an employed cohort in the same proportion — simply because a person usually has to be healthy enough to hold a job in the first place. As a result, general-population mortality is usually higher than that of any occupational cohort, even an unexposed one.

Regional differences in social, behavioral, nutritional, and environmental factors can also make a regionally based cohort's mortality differ from a nationwide general population — for reasons that have nothing to do with the exposure under study.

Bottom line: the SMR is a potentially biased estimate of any risk factor's true effect.

EXTERNAL COMPARISONS

Softening the Bias, or Trading It Away

Choose a closer reference

Compare against a reference population as similar as possible to the cohort — except for the risk factor of interest — rather than the general population.

Compare cohort to cohort

Trade a smaller bias for a larger variance: the reference is now itself a sample, so confidence intervals must be built with genuinely internal methods, and they widen.

The choice is a trade-off between accepting a (often small) bias and accepting the lower statistical power of a wider confidence interval. Some methods even blend external reference rates into an internal comparison to stabilize it. Where possible, the chapter's advice is simple: try both.

MEASURING AN EFFECT

Comparing Exposed to Unexposed, Directly

The main goal of internal comparison is simple: measure morbidity or mortality in the cohort's own exposed subjects against its own unexposed subjects, and look for a dose-response relationship across exposure categories.

Risk ratio (RR)

Risk among the exposed $\div$ risk among the unexposed. Also called relative risk — the most popular measure of association.

Incidence ratio (IR)

Incidence rate among the exposed $\div$ incidence rate among the unexposed — the person-time-based counterpart of the risk ratio.

MEASURING AN EFFECT

The Attributable-Risk Family

Ratios describe relative strength; this family measures absolute, real-world impact instead.

Attributable risk (AR)

Risk (exposed) minus risk (unexposed): the absolute extra risk from exposure. (A distinct, easily confused sense of “absolute risk.”)

Attributable fraction (AF)

$AR \div \text{risk among the exposed}$: the share of an exposed person’s own risk due to the exposure. Useful for counseling individuals.

Population attributable risk (PAR)

$AR \div \text{prevalence of exposure}$: the absolute population risk due to exposure. Estimates public-health impact.

Population attributable fraction (PAF)

The share of **all** disease events in the population due to the exposure — what could be prevented if it were eliminated.

MODELING CHANGING EXPOSURE

The Trouble With a Single Snapshot

Many exposures accumulate continuously over follow-up — cumulative years of smoking, or a cumulative occupational dose. Labeling each person by their **total** exposure at the end of the study, and treating that as their exposure the whole time, quietly distorts the result.

Person-time accrued **early** in follow-up — when someone's true cumulative exposure was still low — gets wrongly credited to whatever high exposure category they eventually reach. The result: risk is overestimated in the low-exposure category, and underestimated in the high-exposure category.

MODELING CHANGING EXPOSURE

The Fix: Slice Each Person's Time Instead

Rather than one fixed label per person, split each person's disease-free follow-up into pieces, and assign each piece to the age-and-exposure category they actually belonged to **at that moment**. Most cohort members end up contributing pieces of person-time to several different cells as they age and accumulate more exposure.

A concrete illustration

A subject followed from age 36 to 53, diagnosed at 41, splits their disease-free time across three cells: age 30–39 with low cumulative exposure, age 30–39 with moderate exposure, and age 40–49 with moderate exposure — the cell they occupied at diagnosis receives the case itself.

When a case occurs, it is counted in whichever cell the person occupied at the moment of diagnosis — not wherever they ended up.

MODELING CHANGING EXPOSURE

Three Further Wrinkles

Varying duration

Exposure can start, stop, and start again — not just accumulate steadily from day one.

Intensity

Exposure can be weighted by how intense it was, in units like ppm-years, so a short intense spell can count the same as a longer, milder one.

Lag

Exposure is sometimes lagged by a fixed number of years, treating the most recent window as unexposed — too soon for it to have plausibly caused disease.

If a confounder like smoking also changes during follow-up, person-time is split further still to track it. The bookkeeping grows complex quickly, which is why specialized software exists to manage it (Coleman et al. 1986; modern packages such as Stata and Epicure lift some older restrictions).

MODELING CHANGING EXPOSURE

The Payoff: One Tidy Table

All that slicing produces something simple: one row per combination of age category, exposure category, and any other stratifying variable, recording only two numbers — the person-years the whole cohort accumulated in that cell, and how many disease events occurred in it.

Age group	Exposure category	Person-years	Cases
30–39	Unexposed	4,210	6
30–39	Low exposure	1,860	4
40–49	Unexposed	3,570	9
40–49	High exposure	920	7

Illustrative figures. This compact format — one line per cell — is exactly what feeds the regression models on the next two slides.

MODELING CHANGING EXPOSURE

Poisson Regression

The standard model for analyzing rates in exactly this kind of table. It assumes the number of events in each cell follows a Poisson distribution, with an expectation set by that cell's person-time multiplied by its true underlying rate.

The model splits the log of the rate into two parts: one adjusting for “nuisance” variables like age and calendar period, and one capturing the exposure category's own effect relative to a baseline category. Exponentiating that exposure effect gives an adjusted rate ratio.

Full modeling strategy goes well beyond this guide — see the Handbook's dedicated regression and survival-analysis chapters, and Breslow and Day (1987, Chap. 4) for a comprehensive account.

MODELING CHANGING EXPOSURE

Cox Regression: Skip the Categories

Turning a continuous variable like exact age into categories always throws away some information — a loss of statistical power. Cox proportional-hazards regression avoids this for the time axis by working directly with the exact moments events occur. One choice still has to be made up front: which variable serves as the underlying time scale — age, calendar time, or time since entry into the cohort — since every risk set to follow is built along whichever scale is chosen.

The key idea: risk sets

At each moment an event occurs, form the “risk set” — everyone still at risk and under observation at that instant — and compare the case’s exposure to the rest of the risk set’s exposure.

This turns out to be mathematically identical to a conditional logistic regression matched on risk set — the bridge to the next idea: nested case-control sampling, which exploits that equivalence directly.

NESTED DESIGNS

Nested Case-Control Sampling

Rather than gathering costly, detailed information — interviews, biomarkers, job histories — on the entire cohort, sample just a small number of controls, typically three or four, from each case's risk set, and collect the detailed data only on the case and those controls.

Why it works

The same conditional-logistic-regression machinery behind Cox regression still yields statistically valid estimates — at a fraction of the data-collection cost.

The drawback

With few controls, the most informative ones — whose exposure genuinely differs from the case's — might be missed by chance, especially for one specific exposure of interest.

NESTED DESIGNS

Countermatching Fixes the Drawback

Langholz and Clayton (1994) proposed deliberately selecting controls who are as **different** as possible from the case on a surrogate measure of exposure, rather than picking controls at random.

Steenland and Deddens (1997) found that countermatching with 3 controls per case gave about a **25% increase in efficiency** over random sampling — roughly the precision of random sampling with 10 controls per case.

Near the precision of a much larger random sample, for the cost of a much smaller one. Later refinements have pushed the efficiency even further (Rivera and Lumley 2016).

A WORKED EXAMPLE

Metal-Working Fluids and Bladder Cancer

Bourgkard et al. (2009) found a significant excess of bladder-cancer mortality among French steel-factory workers exposed to metal-working fluids (MWF), identified with a job-exposure matrix. Colin et al. (2018) extended the cohort to more factories and more follow-up time, then studied bladder-cancer incidence with a nested, countermatched case-control design.

Design

3 controls per case, matched on age and sex, countermatched across 4 surrogate exposure strata from the job-exposure matrix.

Data

Each case and control interviewed face to face for detailed occupational history and confounders — far too costly cohort-wide.

Result

Confirmed the cohort finding: a significant increase in risk with duration of exposure to straight (neat) fluids; a nonsignificant increase for water-based fluids.

Because of the countermatched sampling, the log of each risk set's stratum sizes was included as an offset in the conditional logistic regression, with matching adjustments to the variance.

SELECTING A POPULATION

Representative, or Just Convenient?

For external comparisons to be valid, a cohort ought to be representative of the general population. In practice, most cohorts are convenience samples, drawn from whatever group happens to be accessible.

Truly representative, census-based cohorts are rare, mostly because of confidentiality protections — though notable exceptions exist: a Swedish occupational census sample, and a 10% sample of the Canadian labor force built from social insurance numbers (Howe and Lindsay 1983).

Occupational cohorts are usually accessible when a company or union itself wants to know whether a suspected hazard is real. Many now-confirmed carcinogens were first flagged this way, often after earlier evidence from animal studies (Tomatis et al. 1990).

SELECTING A POPULATION

Volunteer Cohorts, and the Pooling Project

Many “multipurpose” cohorts recruit volunteers: the Nurses’ Health Study (Willett et al. 1992), a Canadian cohort drawn from a mammography-screening trial (Howe et al. 1991), a Swedish screening-program cohort (Wolk et al. 1998), and EPIC, a ten-country European cohort (Riboli and Kaaks 1997). Response was good in most countries, but volunteers skew more health-conscious and more educated than the general population.

The Pooling Project combines many such cohorts into one “megastudy,” detecting risks no single cohort could show alone — originally for diet and breast cancer (Smith-Warner et al. 2006), since extended to lung cancer and other diseases. OMEGA-NET pursues a similar goal for occupational cohorts.

A KEY REASSURANCE

Internal Validity Does Not Need Representativeness

A truly representative cohort is essentially unobtainable — the very act of volunteering, and being free to refuse, is itself a form of selection. So comparing a volunteer cohort's overall rates to the general population may well be invalid.

But provided the recruitment mechanism is unbiased with respect to the **exposure of interest specifically**, and the data allow investigators to stratify into exposed and unexposed subgroups, the estimated exposure–outcome association stays valid. External validity can fail while internal validity — the thing that actually answers the question — holds.

In practice, investigators still work to protect that mechanism — minimizing nonparticipation wherever possible, and where some remains, comparing nonparticipants to participants on whatever characteristics are available to check for the selection this slide describes.

SELECTING A POPULATION

A Worked Example: What Selection Distorts, and What It Doesn't

Estimate	Target population (truth)	Study population, fair sampling	Study population, oversampled exposed
Prevalence of disease	10%	10%	16%
Prevalence of risk factor	20%	20%	71%
Relative risk	2.67	2.67	2.67
Odds ratio	3.08	3.08	3.08
Attributable risk	125/1,000	125/1,000	125/1,000

Oversampling the exposed badly distorts the raw prevalences (orange) — but as long as selection depends only on exposure, not on exposure and disease jointly, the measures of association (green) stay exactly right. Selection bias only threatens these once participation differs by both exposure and disease together, or diseased individuals are specifically overrepresented. One exception: because PAR and PAF are built from exposure prevalence, they inherit that same distortion — so they can still be wrong even when RR, OR, and attributable risk remain correct.

EXPOSURE AND CONFOUNDING

Measuring Exposure at Scale

Some cohorts, especially occupational ones, inherit exposure data from routine records kept for other purposes — a real strength, since exposure was documented before disease occurred. Population-based cohorts have no such luxury: investigators must collect exposure data specifically, and at cohort scale that almost always means a mailed, self-administered questionnaire rather than an interview.

A self-administered questionnaire can only capture so much detail, and respondents can only recall so accurately — a problem historically underappreciated, and probably part of why so many diet-focused cohort studies in particular have produced negative results. Even smoking history cannot be regarded as precisely measured.

EXPOSURE AND CONFOUNDING

Two Kinds of Misclassification

Misclassification is **differential** when its degree depends on the person's outcome status, and **nondifferential** when it does not. Cohort studies are typically nondifferential, since exposure is measured at baseline, before any future outcome is known — unlike case-control studies, where differential misclassification is a far bigger concern.

A worked example: a cohort study of smoking and myocardial infarction, with a realistic 10% nondifferential misclassification rate in classifying smokers.

True rate ratio	Observed rate ratio
2.00	1.83

Nondifferential misclassification of a dichotomous exposure always biases toward the null, and more misclassification means more bias. (With more than two exposure categories, the bias can run either toward or away from the null.)

EXPOSURE AND CONFOUNDING

Exposure That Changes Over a Lifetime

Case-control studies can reconstruct a detailed exposure biography — changes in intensity, quitting dates — from one interview, at the cost of recall uncertainty. A concurrent cohort can genuinely track changes as they happen, but only if the design specifically re-administers the exposure questionnaire on a regular schedule.

The Nurses' Health Study (Willett et al. 1992) solved this by treating the period between the first and second exposure surveys, and the period after the second survey, as two separate “cohorts” — statistically justified because different blocks of person-time are independent, regardless of how much they share the same people.

Most analyses, though, simply use exposure as measured at baseline — and for cancers with a long induction period, that distance has not always been treated as a major disadvantage.

EXPOSURE AND CONFOUNDING

Toward More Objective Exposure Measures

Film badges

Radiation workers can have cumulative, and sometimes peak, exposure measured directly rather than estimated.

Job-exposure matrix

Job titles as rows, agents of interest as columns, each cell holding the probability or intensity of exposure — often built from historical hygiene measurements.

Biological specimens

From simple blood pressure and cholesterol to today's biomarkers of exposure and genetic factors, as in EPIC.

A job-exposure matrix, extended with portable electromagnetic-field meters and historical usage records, helped electric-utility studies identify strong links between high electric-field exposure and leukemia and non-Hodgkin's lymphoma (Theriault et al. 1994; Miller et al. 1996; Villeneuve et al. 1998).

EXPOSURE AND CONFOUNDING

The Duration Trap

When true cumulative dose cannot be measured directly, duration of exposure is often used as a stand-in. That substitution is dangerous whenever the exposure has an early, toxic effect: affected workers tend to change jobs quickly, so they never accumulate a long recorded duration — even though their true dose, and their risk, may already be substantial.

If those early-affected workers are also a genetically susceptible subgroup, a brief exposure in them can produce the same incidence as a much longer exposure in a less susceptible subgroup — and the resulting absence of a clean dose-response pattern can be wrongly read as evidence the exposure is not causal at all (Blair and Stewart 1992).

EXPOSURE AND CONFOUNDING

Confounding: Still the Persistent Challenge

Historical cohorts especially may never have captured all the relevant confounders, and even confounder data that was collected may lack the precision needed to fully account for its effect.

A live example: does fruit and vegetable consumption protect against lung cancer, independent of smoking? Two large, well-regarded cohort analyses fully adjusted for smoking, in their own investigators' judgment (Miller et al. 2004; Smith-Warner et al. 2003) — yet an IARC working group was not convinced residual confounding by smoking had been ruled out, and judged the evidence merely “limited,” not “sufficient” (IARC 2003).

Even careful, well-resourced studies can leave genuine, unresolved disagreement about whether confounding has truly been handled.

EXPOSURE AND CONFOUNDING

The Healthy Worker Survivor Effect

Different from the Healthy Worker Effect discussed earlier. This is a selection effect **within** the workforce over time: the healthiest workers accrue the most exposure, because workers in poorer health reduce their hours, move to lower-exposed jobs, or leave employment entirely — creating a downward bias in the apparent dose-response relationship.

Brown et al. (2017) describe modern “g-methods” that can, unlike standard regression adjustment, estimate the unbiased causal effect of a hypothetical intervention that lowers everyone’s exposure.

A re-analysis of a European cohort of titanium-dioxide workers (Guseva Canu et al. 2022) modeled year-by-year health, employment, and exposure as a connected chain, and found a statistically significant effect of health status on the next year’s exposure — direct evidence the effect was operating in this real cohort.

DETERMINING OUTCOMES

Rare Diseases Need Big Cohorts and Long Follow-Up

Most diseases are genuinely rare, with rates typically expressed per 100,000 people. Accumulating enough cases to say anything meaningful requires a large population, a long follow-up, or both — and many diseases, especially cancers, also carry a long induction period, often exceeding 10 years, sometimes 20.

The British Doctors Study found much higher lung-cancer risks from smoking after 40 years of follow-up than its own earlier 10- and 20-year reports had shown (Doll et al. 1994b) — apparently because it is the **duration** of smoking, more than its intensity, that dominates the risk, and that effect only fully reveals itself after decades.

DETERMINING OUTCOMES

A Major Advantage: Protection Against Outcome-Dependent Recall Bias

Because exposure is measured before anyone knows who will get the disease, cohort studies are protected against a major form of information bias that haunts case-control studies: people who already have a disease may recall or report their past exposures differently, simply because they are searching for an explanation. This does not mean cohort exposure measurement is error-free — only that it cannot be distorted by knowledge of an outcome that has not happened yet.

This exposure-before-outcome ordering is arguably the cohort design's single most valuable structural property. It also means one cohort can identify multiple distinct endpoints at once — several cancer sites, or several types of cardiovascular disease — from one underlying study population.

DETERMINING OUTCOMES

Ascertainment Bias

Because cohort members know they are being studied, and are often under periodic study-related contact or testing, the very fact of being in a study can lead to more intensive diagnostic effort — and so a higher apparent diagnosis rate — than an equivalent person outside the study would receive.

Special surveillance stays valid for **internal** comparisons, as long as it is applied equally to exposed and unexposed members. That same surveillance would invalidate an **external** comparison against general-population rates, which are not subject to it.

A WORKED EXAMPLE

Mitral Valve Prolapse and Stroke

Orencia et al. (1995) followed a Mayo Clinic cohort of people diagnosed with mitral valve prolapse (MVP) for stroke, and compared their rate to the general population of Olmsted County, Minnesota.

Overall SMR

2.1

double the stroke risk

But MVP can be diagnosed by ordinary auscultation, **or** serendipitously during an echocardiogram ordered for another reason — a heart attack, heart failure, atrial fibrillation — conditions already linked to stroke risk on their own.

Splitting the cohort by diagnostic method told the real story: the auscultation-diagnosed group showed **no** increased stroke risk at all. The entire elevated risk was confined to the serendipitously diagnosed group — a textbook case of ascertainment bias masquerading as a disease association.

DETERMINING OUTCOMES

Keeping Track, and What It Costs

Many cohorts maintain contact through periodic mail, reaching the member directly or their designated physician, to identify a study endpoint or death from any cause. It works — but it is costly, risks losing a growing share of the cohort over time, and family members are not always willing to help once a participant has died.

Losses to follow-up always cost statistical power. Whether they also cost **validity** depends entirely on why people were lost — the question the next slide answers.

DETERMINING OUTCOMES

When Losses to Follow-Up Bias the Results

Losses that do not differ by exposure or outcome cost power, but no bias. Losses that differ by exposure alone leave the ratios below unbiased, just like oversampling did earlier. The genuinely dangerous patterns:

Estimate	True value	Losses differ by outcome	Losses differ by both
Relative risk	2.67	1.60	1.48
Odds ratio	3.08	3.08	1.54
Attributable risk	125/1,000	260/1,000	36/1,000

When loss to follow-up depends on the outcome itself — alone, or combined with exposure — the bias can push estimates in genuinely unpredictable directions.

DETERMINING OUTCOMES

Record Linkage

Beyond direct contact, outcomes can come from medical records, hospitals, vital statistics systems, and disease registries. Determining whether a record belongs to a specific cohort member is called record linkage — increasingly done by computer, especially efficient when a shared national identifying number is available.

This makes recruitment-time data collection matter enormously: full name, date and place of birth, mother's maiden name, an identifying number, current address — and the names of friends or relatives, to help trace someone later if other methods fail.

National resources include the NHS Central Register (UK), the Canadian National Mortality Database, the National Death Index (US), and Scandinavian population registers; newer genetic biobanks, like Estonia's, are joining them. In Canada, data returned through national linkage is generally anonymized unless the subject specifically consented to linkage.

ETHICS

Consent for a Purpose Not Yet Known

Informed consent, developed for clinical trials, has generally extended to cohort studies. But biological samples in multipurpose cohorts raise a distinctive problem: at the time specimens are collected, their precise future use is often genuinely unknown.

Most EPIC participants gave blood in the early 1990s, before genetic studies were common — so their consent forms rarely mentioned genetics. Years later, that gap complicated approval for genetic substudies. The proposed fix, “broad consent” (Sandu and Frunza 2019), lets a participant agree upfront to future, not-yet-specified research, though acceptance varies by country.

ETHICS

Should Participants Learn Their Own Results?

A genuinely unresolved question, with two contrasting traditions:

The “European” view

Results may be impossible to interpret individually until the whole research program concludes, so it may be unnecessary, even unethical, to share them early.

The “American” view

It is ethically inappropriate for investigators to unilaterally decide, on someone else’s behalf, whether they get to know something about themselves.

Genetic results especially may arrive too late to help the original subject, yet still matter to their relatives — and such knowledge is not always a blessing. The one point of agreement: information of real potential benefit should be shared.

ETHICS

Historical Cohorts Raise Fewer Concerns

Consent for data that already exists, collected in the past for some other purpose, generally does not arise the way it does for prospective data collection — though real questions can remain about informing surviving subjects of a study’s eventual findings.

Because this kind of research is unlikely to harm participants and confidentiality can typically be well maintained, ethics committees generally approve such studies without major difficulty — and record linkage for outcomes, discussed earlier, maintains that same confidentiality.

WRAPPING UP

What a Trustworthy Cohort Study Needs

Investigators skilled in the peculiarities of planning and conducting a cohort study

A sensible source for recruitment, and hypotheses worth evaluating

A validated questionnaire, administered without bias, for enrollment and follow-up alike

Quality-controlled procedures for collecting biological material, where relevant

Facilities for data entry

Expertise and facilities for analysis and interpretation

WRAPPING UP

Cohort Studies vs. Case-Control Studies

Cohort studies are a critical method for evaluating causality in epidemiology, and can also be used to evaluate screening programs. Formal evidence hierarchies usually classify both cohort and case-control studies as “level II” evidence, below the “level I” of randomized trials — though cohort studies are often ranked somewhat higher within that tier, largely because case-control studies are more susceptible to recall bias.

An easily missed nuance: cohort studies are not simply “better” in every way. They carry their own particular weakness — a greater propensity for measurement error than case-control studies, which can sometimes afford more intensive measurement of a smaller number of subjects.

Both designs, despite their different weaknesses, continue to play an essential role in disease epidemiology.

BOTTOM LINE

Watch, Don't Ask

Classify a healthy population by exposure, then let time reveal who becomes ill and who does not. That simple structure — exposure measured first, disease discovered later — is what gives cohort studies their unmatched protection against recall bias, and their power to trace a single exposure's fingerprints across many diseases at once.

What it costs in return: size, patience, and expense — and a demanding list of ways that selection, misclassification, confounding, and loss to follow-up can each quietly bias the very comparison the design exists to protect.

From Snow's cholera map to today's million-person biobanks, the logic has not changed — only the scale of the population willing to be watched.

SOURCE & ATTRIBUTION

Wild, P., Miller, A. B., Goff, D. C. Jr., & Bammann, K. (2023). “Cohort Studies.”

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